

① Assignment 4 [MA725]

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Ans.

$$\begin{aligned}x_1 - x_3 &= 0.2, \\ -\frac{1}{2}x_1 + x_2 - \frac{1}{4}x_3 &= -1.425, \\ x_1 - \frac{1}{2}x_2 + x_3 &= 2\end{aligned}$$

with solution $(0.9, -0.8, 0.7)^T$.

② Strict Diagonal Dominance :-

A matrix is sdd if $|a_{ii}| > \sum_{j \neq i} |a_{ij}| \forall i$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -0.5 & 1 & -0.25 \\ 1 & -0.5 & 1 \end{bmatrix}$$

$$r_1: |1| > |0| + |-1| \quad \times$$

$$r_2: |1| > |-0.5| + |-0.25| \quad \checkmark$$

$$r_3: |1| > |1| + |-0.5| \quad \times$$

It fails for r_1 & $r_3 \therefore A$ is not strictly diagonally dominant.

③ Spectral radius of T_g :-

$$T_g = (D - L)^{-1} U$$

$$D - L = \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ 1 & -0.5 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -0.25 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(D - L)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ -0.75 & 0.5 & 1 \end{bmatrix}$$

$$T_g = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ -0.75 & 0.5 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -0.25 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -0.75 \\ 0 & 0 & -0.625 \end{bmatrix}$$

$$\lambda = 0, 0, -0.625$$

$$\therefore \rho(T_g) = 0.625$$

④ Gauss-Seidel iteration :-

$$\text{initial guess, } x^{(0)} = [0 \ 0 \ 0]^T$$

(2) Iteratives:

$$x_1^{(k+1)} = 0.2 + x_3^{(k)}$$

$$x_2^{(k+1)} = -1.425 + 0.5x_1^{(k+1)} + 0.25x_3^{(k)}$$

$$x_3^{(k+1)} = 2 - x_1^{(k+1)} + 0.5x_2^{(k+1)}$$

Iteration 1

$$x_1^{(1)} = 0.2 + 0 = 0.2$$

$$x_2^{(1)} = -1.425 + 0.5(0.2) + 0.25(0) = -1.425 + 0.1 = -1.325$$

$$x_3^{(1)} = 2 - (0.2) + 0.5(-1.325) = 1.1375$$

$$x^{(1)} = [0.2 \quad -1.325 \quad 1.1375]^T$$

Iteration 2

$$x_1^{(2)} = 0.2 + 1.1375 = 1.3375$$

$$x_2^{(2)} = -1.425 + 0.5(1.3375) + 0.25(1.1375) = -0.4718$$

$$x_3^{(2)} = 2 - (1.3375) + 0.5(-0.4718) = 0.4265$$

$$x^{(2)} = [1.3375 \quad -0.4718 \quad 0.4265]^T$$

Iteration 3

$$x_1^{(3)} = 0.2 + 0.4265 = 0.6265$$

$$x_2^{(3)} = -1.425 + 0.5(0.6265) + 0.25(0.4265) = -1.0050$$

$$x_3^{(3)} = 2 - (0.6265) + 0.5(-1.0050) = 0.87089$$

We continue this iteration process until $\|x^{(k)} - x^{(k-1)}\|_\infty < 0.01$. with true soln. $(0.9, -0.8, 0.9)$.

(d) Updated equations:-

Let A' after modification is, $A' = \begin{bmatrix} 1 & 0 & -2 \\ -0.5 & 1 & -0.25 \\ 1 & -0.5 & 1 \end{bmatrix}$

$$A' = D - L - U, \quad D - L = \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ 1 & -0.5 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0.25 \\ 0 & 0 & 0 \end{bmatrix}$$

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$$T_g = (2 - L)^{-1} U$$

$$(2 - L)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ -0.75 & 0.5 & 1 \end{bmatrix}$$

$$T_g = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ -0.75 & 0.5 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0.25 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 1.25 \\ 0 & 0 & -1.375 \end{bmatrix}$$

$$\det \begin{bmatrix} -\lambda & 0 & 2 \\ 0 & -\lambda & 1.25 \\ 0 & 0 & -1.375 - \lambda \end{bmatrix} = 0 \quad \lambda = 0, 0, -1.375$$

$$\rho(T_g) = |\lambda_{\max}| = 1.375$$

$\therefore \rho(T_g) = 1.375 > 1$, this method is guaranteed to diverge. thus with this change the iterative values will not reach the soln. (0.9, -0.8, 0.7).

Ans 2. SOR :-

$$w = 1.06, \quad A = \begin{bmatrix} 5 & 1 & -2 \\ 1 & 10 & 2 \\ -2 & 2 & 10 \end{bmatrix}, \quad b = \begin{bmatrix} -1.5 \\ 0.2 \\ -1.0 \end{bmatrix}$$

$$Ax = b$$

using Jacobi & Gauss-Seidel methods.

$$x^{(0)} = [0, 0, 0]^T$$

SOR formulae

$$x_1^{(k+1)} = (1 - 1.06) x_1^{(k)} + \frac{1.06}{5} [-1.5 - 2x_2^{(k)} + 2x_3^{(k)}]$$

$$x_2^{(k+1)} = (1 - 1.06) x_2^{(k)} + \frac{1.06}{10} [0.2 - x_1^{(k+1)} - 2x_3^{(k)}]$$

$$x_3^{(k+1)} = (1 - 1.06) x_3^{(k)} + \frac{1.06}{10} [-1.0 + 2x_1^{(k+1)} - 2x_2^{(k+1)}]$$

$$\text{Step 1: } x_1^{(1)} = (1 - 1.06)(0) + \frac{1.06}{5} [-1.5 - 0 + 0] = -0.318$$

$$x_2^{(1)} = (1 - 1.06)(0) + 1.06/10 [0.2 - (-0.318) - 0] = 0.0549$$

$$x_3^{(1)} = (1 - 1.06)(0) + 1.06/10 [-1.0 + 2(-0.318) - 2(0.0549)] = -0.1850$$

(4) Step 2:

$$x_1^{(2)} = (1 - 0.06)(-0.318) + \frac{1.06}{5} [-1.5 - 0.0549 + 2(-0.1850)]$$
$$= -0.3890$$

$$x_2^{(2)} = (-0.06)(0.0549) + \frac{1.06}{10} [0.2 - (-0.389) - 2(-0.185)]$$
$$= 0.0983$$

$$x_3^{(2)} = (-0.06)(-0.185056) + \frac{1.06}{10} [-1.0 + 2(-0.3890) - 2(0.0983)]$$
$$= -0.1982$$

Step 3:

$$x_1^{(3)} = (-0.06)(-0.389) + \frac{1.06}{5} [-1.5 - 0.0983 + 2(-0.1982)]$$
$$= -0.399550$$

$$x_2^{(3)} = (-0.06)(0.0983) + \frac{1.06}{10} [0.2 - (-0.3995) - 2(-0.1982)]$$
$$= 0.0996$$

$$x_3^{(3)} = (-0.06)(-0.1982) + \frac{1.06}{10} [-1.0 + 2(-0.3995) - 2(0.0996)]$$
$$= -0.1999$$

Step 4:

$$x_1^{(4)} = (-0.06)(-0.39955) + \frac{1.06}{5} [-1.5 - 0.0996 + 2(-0.1999)]$$
$$= -0.3999$$

$$x_2^{(4)} = (-0.06)(0.0996) + \frac{1.06}{10} [0.2 - (-0.3999) - 2(-0.1999)]$$
$$= 0.0999$$

$$x_3^{(4)} = (-0.06)(-0.1999) + \frac{1.06}{10} [-1.0 + 2(-0.3999) - 2(0.0999)]$$
$$= -0.1999$$

	Step 1	Step 2	Step 3	Step 4
x_1	-0.318	-0.3890	-0.3995	-0.3999
x_2	+0.0549	0.0983	0.0996	0.0999
x_3	-0.1850	-0.1982	-0.1999	-0.1999